



English astronomer John Flamsteed's meticulous observations of the night sky formed the basis of the first modern star catalog.

although **spiders do make silk**, it is much thinner thread, and he argued that spiders were too aggressive to use commercially.

Mathematician John Keill (1671–1721) published a paper claiming that Gottfried Leibniz, a German mathematician, stole the idea of calculus from British mathematician and physicist Isaac Newton. It is now thought that both men independently developed the **basis of calculus**.

The following year in Italy, Bolognese nobleman **Luigi Fernando Marsili** asserted that **corals** are plants, not animals. His mistaken view prevailed at the time, although others had realized that corals are animals.



#### Spider web

In 1710, Frenchman René de Réaumur showed that spiders produce silk. Spiders use the thread to make webs to catch prey or as cocoons for their young.

**THE FIRST FULL-SCALE STEAM ENGINE**, built by English engineer **Thomas Savery** in 1698, proved too dangerous for general use because high pressure in its boiler tended to cause explosions. But English ironmonger **Thomas Newcomen** (1663–1729) overcame the danger in 1712 to create the world's **first practical steam engine**. Newcomen's solution was to boil water in an isolated chamber and send the steam into a cylinder with a piston at low pressure. When steam flowed into the cylinder, it pushed the piston up. A valve closed, cold water was sprayed in, and the steam condensed, creating a vacuum that pulled the piston down, moving the engine's beam. Newcomen's steam engine was so successful that soon thousands of them were installed in mines across Britain and Europe to pump out floodwater.

In London this year, **Isaac Newton** and astronomer **Edmond Halley** enraged British astronomer **John Flamsteed** by publishing a catalog of more than 3,000 stars based on Flamsteed's observations made over 40 years at the Royal Greenwich Observatory. Newton and Halley believed the data should be published, but Flamsteed felt it was not thorough enough. He was so incensed by the

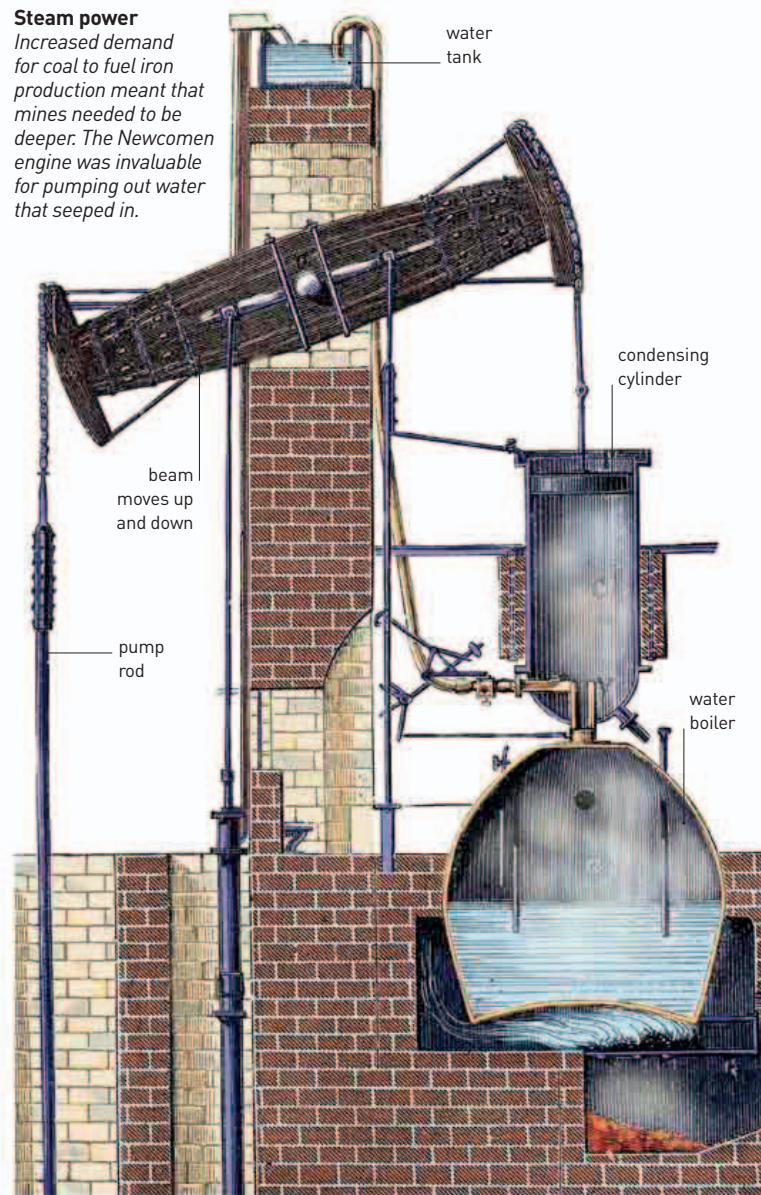
## 10 GALLONS THE AMOUNT OF WATER PUMPED EACH MINUTE BY NEWCOMEN'S FIRST ENGINE

publication that he gathered and burned 300 of the 400 printed copies.

Swiss mathematician **Jacob Bernoulli's** book *Ars Conjectandi* (*The Art of Conjecture*) was published seven years after his death, in 1713. It introduced the **Law of Large Numbers**, which says that the more times you perform an experiment, the closer the average result tends to be to the average of a large number of experiments. That year, Bernoulli's nephew, **Nicolas Bernoulli**, devised the **St. Petersburg paradox** familiar to probability theorists today. It is based on a theoretical lottery game that seems to allow an infinite win, yet it is one that nobody with any sense would enter.

#### Steam power

Increased demand for coal to fuel iron production meant that mines needed to be deeper. The Newcomen engine was invaluable for pumping out water that seeped in.



1711 English clergyman Stephen Hales measures animal blood pressure

1712 Thomas Newcomen invents the first practical steam engine

1712 John Flamsteed's star catalogue is published

1713 Johann Bernoulli's *Ars Conjectandi* is published posthumously

September 9, 1713 Swiss mathematician Nicolas Bernoulli outlines the St. Petersburg paradox